TANMAY HINGE

www.tanmayhinge.com | hingetanmay@gmail.com
Mumbai, Maharashtra, India
Available for doctoral positions starting January 2027

EDUCATION

M.S. Computer Science, California State University, Chico, USA 2022 to May 2024

- **GPA 3.342** over 39 units, taught and examined in English.
  Coursework: Applied Machine Learning, Applied Computer Vision, Applied Graph Theory, Algorithms and Computability, Advanced Computer Security, Research Methods in Computer Science.
- **Capstone -** Statistical Survey for assessing candidate employability on features from two surveys

B.E. Computer Engineering, Pillai College of Engineering, University of Mumbai, India 2018 to June 2022

- **CGPI 7.83** out of 10.00.
- Coursework: Digital Signal and Image Processing, Machine Learning, Artificial Intelligence, NLP and HCI.

RESEARCH STUDIES

Drift of CA1 Context Codes Across Days (code and report) 2026

- **Data.** Public hippocampal CA1 miniscope recordings (Zaki et al.): 2 mice, six days, three rooms, streamed over the network.
- **Method.** Wrote a cell matcher that aligns fields of view and pairs cells by outline overlap. On the mouse that has the authors' CellReg tables it agrees on 98.5 to 100 percent of pairs. It followed 223 cells across all six days.
- **Result.** Activity was more alike across days than chance, but no more alike for the same room than for different rooms. Similarity fell as the gap between sessions grew. A room decoder scored 0.497 on unseen days, where 0.5 is guessing. Freezing and movement do not explain it.
- **Limit.** With 2 mice this is a small case study.

Probing a frozen rPPG encoder for identity under input degradation (code and report) 2026

- **Method.** Froze two published camera-based heart-rate models and fitted linear probes to their representation. Every number is set against four floors, one an untrained model of the same architecture. 107 tests.
- **Result.** Identity of 120 people decoded at 100 percent, where guessing gives 0.83 percent, at every blur level. It broke only when the face-shaped outline was deleted too, so identity sat in the silhouette.
- **Limit.** Pulse was faint: heart-rate error 25.3 bpm against 28.8 for ignoring the video, not separable from that floor once bootstrapped. The report says so.

Predicting treatment delays of more than 90 days in prostate cancer (code and report) 2026

- **Data.** 330,827 men in the US SEER registries, 2010 to 2022. The share waiting more than 90 days rose from 36.0 to 52.3 percent.
- **Method.** Penalised logistic regression and LightGBM, feature blocks added in a fixed order, bootstrap intervals throughout. Public protocol with an amendment log. 250 tests.
- **Result.** Area and marital status add a small amount over the clinical record. AUC 0.592 to 0.663, where 0.5 is guessing.

EXPERIENCE

Visiting Researcher, AI for Conservation (Elephant Listening Project) October 2026 to present
Sensor Fusion Research Group, California State University, Chico

Computer Science Faculty, Inver Hills Community College, Minnesota, USA July 2024 to June 2026

- Designed and taught the full Applied AI certificate programme, from course sequence and prerequisites to teaching material and assessment.
- Taught four core courses (introductory computing, C++, Python, Java). Mentored more than 1,000 students over two years. Ran outreach to five local high schools.

AI Researcher, Regace AI February 2026 to present

- Retrieval engine over regulated text, with structure-aware ingestion, hybrid retrieval with reranking and citation-grounded generation, so every claim traces to a source passage. In production over 10,000 indexed documents with two pilot clients.

Computer Science Tutor and Laboratory Assistant, CSU Chico August 2023 to May 2024

- Lab assistant for Programming and Algorithms II and Data Structures and Algorithms. Tutored C++, algorithms and computer security.

Software Engineer Intern, VettX Inc., California, USA May 2023 to August 2023

- Built a client-onboarding platform in React with the Vidyard API, and an assisted Chrome extension that shortened seller outreach by roughly 40 percent.

TECHNICAL SKILLS

Languages	Python, C++, Rust, Java, TypeScript
Machine learning	PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, LightGBM, MLflow
Data and simulation	NumPy, SciPy, pytest, Minian, CellReg, ezTrack, NWB and DANDI, MuJoCo, Isaac Gym
Infrastructure	Docker, Kubernetes, CI/CD, Git, AWS, GCP, Azure, PostgreSQL, Linux, LaTeX

LANGUAGES

English, native and fluent. Marathi, native and fluent. German, beginner.

REFEREES

All three have agreed to be contacted directly.

Samuel B. Siewert, Professor of Computer Science, California State University, Chico
sbsiewert@csuchico.edu · Graduate instructor, Applied Computer Vision

Richard C. Tillquist, Assistant Professor of Computer Science, California State University, Chico
rctillquist@csuchico.edu · Graduate instructor, Applied Graph Theory

Tyson R. Henry, Professor of Computer Science (retired), California State University, Chico
TRHenry@csuchico.edu · +1 (530) 828-6174 · Department Chair, and supervisor for my tutoring and student assistant roles.